



De La Salle – College of Saint Benilde
8 Sen. L. Sumulong Memorial Circle, Antipolo, 1870 Rizal
(+63) 8667-1334 / (+63) 8667 – 1300 / 09565576301
<https://www.benilde.edu.ph/antipolo>

DATA STRUCTURE AND ALGORITHMS

Name: Mandal, Ella Mae C.	Date: September 3, 2026
Section: BSIS02	Score:

1. Student Information

```
using System;
class Program
{
    static void Main()
    {
        Console.WriteLine("Enter your name: ");
        string name = Console.ReadLine();

        Console.WriteLine("Enter your course: ");
        string course = Console.ReadLine();

        Console.WriteLine("Enter your year level: ");
        int yearlevel = Convert.ToInt32(Console.ReadLine());

        Console.WriteLine("Enter your section: ");
        string section = Console.ReadLine();

        Console.WriteLine("\n===== STUDENT INFORMATION =====");
        Console.WriteLine("NAME: " + name);
        Console.WriteLine("COURSE: " + course);
        Console.WriteLine("YEAR LEVEL: " + yearlevel);
        Console.WriteLine("SECTION: " + section);
        Console.WriteLine("=====");
    }
}
```

Enter your name: Ella Mae C. Mandal
Enter your course: BSIS
Enter your year level: 2
Enter your section: BSIS02

===== STUDENT INFORMATION =====
NAME: Ella Mae C. Mandal
COURSE: BSIS
YEAR LEVEL: 2
SECTION: BSIS02
=====

C:\Csharp_workspace\example_app\MandalApp\MandalApp\bin\Debug\net10.0\MandalApp.exe (process 15748) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .



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DATA STRUCTURE AND ALGORITHMS

2. Basic Calculator

```
using System;

class Program
{
    static void Main()
    {
        Console.WriteLine("Enter first number:");
        int num1 = Convert.ToInt32(Console.ReadLine());

        Console.WriteLine("Enter second number:");
        int num2 = Convert.ToInt32(Console.ReadLine());

        double sum = num1 + num2;
        double difference = num1 - num2;
        double product = num1 * num2;
        double quotient = num1 / num2;

        Console.WriteLine("==== RESULTS ===");
        Console.WriteLine("Addition: " + (num1 + num2));
        Console.WriteLine("Subtraction: " + (num1 - num2));
        Console.WriteLine("Multiplication: " + (num1 * num2));
        Console.WriteLine("Division: " + (num1 / num2));
    }
}
```

Output:

```
Enter first number: 32
Enter second number: 20

==== RESULTS ====
Addition: 52
Subtraction: 12
Multiplication: 640
Division: 1.6
```

3. Area of a Rectangle

```
using System;

class Program
{
    static void Main()
    {
        Console.WriteLine("Enter length:");
        int length = Convert.ToInt32(Console.ReadLine());

        Console.WriteLine("Enter width:");
        int width = Convert.ToInt32(Console.ReadLine());

        double area = length * width;

        Console.WriteLine("==== Area of a Rectangle ===");
        Console.WriteLine("Area: " + (length * width));
    }
}
```

Output:

```
Enter length: 12
Enter width: 25

==== Area of a Rectangle ====
Area: 30
```



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DATA STRUCTURE AND ALGORITHMS

4. Student Grade

The screenshot displays the Visual Studio IDE with a C# program named 'MandalApp'. The code is as follows:

```
1 using System;
2
3 namespace
4 {
5     static void Main()
6     {
7         Console.WriteLine("Enter First Grade: ");
8         int Grade1 = Convert.ToInt32(Console.ReadLine());
9
10        Console.WriteLine("Enter Second Grade: ");
11        int Grade2 = Convert.ToInt32(Console.ReadLine());
12
13        Console.WriteLine("Enter Third Grade: ");
14        int Grade3 = Convert.ToInt32(Console.ReadLine());
15
16        double average = (Grade1 + Grade2 + Grade3) / 3;
17
18        Console.WriteLine("==== Student Grade ===");
19        Console.WriteLine("Average: " + ((Grade1 + Grade2 + Grade3) / 3));
20    }
21 }
```

The output window shows the program's execution:

```
Enter First Grade: 95
Enter Second Grade: 97
Enter Third Grade: 98
==== Student Grade ====
Average: 96
C:\Csharp_workspace\example_app\MandalApp\MandalApp\bin\Debug\net10.0\MandalApp.exe (process 3540) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

5. Boolean Input

The screenshot displays the Visual Studio IDE with a C# program named 'MandalApp'. The code is as follows:

```
1 using System;
2
3 namespace
4 {
5     static void Main()
6     {
7         Console.WriteLine("Are you a student? (true/false): ");
8         bool IsStudent = Convert.ToBoolean(Console.ReadLine());
9
10        Console.WriteLine("Student Status: " + IsStudent);
11    }
12 }
```

The output window shows the program's execution:

```
Are you a student? (true/false): true
Student Status: True
C:\Csharp_workspace\example_app\MandalApp\MandalApp\bin\Debug\net10.0\MandalApp.exe (process 19452) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```



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DATA STRUCTURE AND ALGORITHMS

6. Age Verification

The screenshot displays the Visual Studio IDE with a C# project named 'MandalApp'. The code in the 'Main' method is as follows:

```
1 using System;
2
3 class Program
4 {
5     static void Main()
6     {
7         Console.WriteLine("Enter your age: ");
8         int age = Convert.ToInt32(Console.ReadLine());
9
10        bool isAdult = age >= 18;
11
12        Console.WriteLine($"Is 18 or older: {isAdult}");
13    }
14 }
```

The output window shows the following text:

```
*MandalApp.exe* (CoreCLR: clrhost): Loaded "C:\Program Files\dotnet\shared\Microsoft.NETCore.App\
*MandalApp.exe* (CoreCLR: clrhost): Loaded "C:\Program Files\dotnet\shared\Microsoft.NETCore.App\
*MandalApp.exe* (CoreCLR: clrhost): Loaded "C:\Program Files\dotnet\shared\Microsoft.NETCore.App\
*MandalApp.exe* (CoreCLR: clrhost): Loaded "C:\Program Files\dotnet\shared\Microsoft.NETCore.App\
The program "[26108] MandalApp.exe" has exited with code 0 (0x0).
```

The debug console shows the user input: "Enter your age: 21" and the program output: "Is 18 or older: True".

7. Full Name

The screenshot displays the Visual Studio IDE with a C# project named 'MandalApp'. The code in the 'Main' method is as follows:

```
1 using System;
2
3 class Program
4 {
5     static void Main()
6     {
7         Console.WriteLine("Enter your first name: ");
8         string firstName = Console.ReadLine();
9
10        Console.WriteLine("Enter your last name: ");
11        string lastName = Console.ReadLine();
12
13        string fullName = firstName + " " + lastName;
14
15        Console.WriteLine($"Full Name: {fullName}");
16    }
17 }
```

The output window shows the following text:

```
*MandalApp.exe* (CoreCLR: clrhost): Loaded "C:\Program Files\dotnet\shared\Microsoft.NETCore.App\
*MandalApp.exe* (CoreCLR: clrhost): Loaded "C:\Program Files\dotnet\shared\Microsoft.NETCore.App\
*MandalApp.exe* (CoreCLR: clrhost): Loaded "C:\Program Files\dotnet\shared\Microsoft.NETCore.App\
*MandalApp.exe* (CoreCLR: clrhost): Loaded "C:\Program Files\dotnet\shared\Microsoft.NETCore.App\
The program "[14156] MandalApp.exe" has exited with code 0 (0x0).
```

The debug console shows the user input: "Enter your first name: Ella Mae" and "Enter your last name: Mandal", followed by the program output: "Full Name: Ella Mae Mandal".



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DATA STRUCTURE AND ALGORITHMS

8. Shopping Calculator

```
using System;
class Program
{
    static void Main()
    {
        Console.WriteLine("Enter product name: ");
        string productName = Console.ReadLine();

        Console.WriteLine("Enter price: ");
        double price = Convert.ToDouble(Console.ReadLine());

        Console.WriteLine("Enter quantity: ");
        int quantity = Convert.ToInt32(Console.ReadLine());

        double total = price * quantity;

        Console.WriteLine($"Product: {productName}");
        Console.WriteLine($"Total Cost: {total}");
    }
}
```

Output:

```
Enter product name: Zebra Sarasa Clip Pen 0.5 Black
Enter price: 89.75
Enter quantity: 3
Product: Zebra Sarasa Clip Pen 0.5 Black
Total Cost: 269.25
```

9. Salary Calculator

```
using System;
class Program
{
    static void Main()
    {
        Console.WriteLine("Enter employee name: ");
        string employeeName = Console.ReadLine();

        Console.WriteLine("Enter hours worked: ");
        double hoursWorked = Convert.ToDouble(Console.ReadLine());

        Console.WriteLine("Enter hourly rate: ");
        double hourlyRate = Convert.ToDouble(Console.ReadLine());

        double salary = hoursWorked * hourlyRate;

        Console.WriteLine($"Employee: {employeeName}");
        Console.WriteLine($"Calculated Salary: {salary}");
    }
}
```

Output:

```
Enter employee name: Ella Mae Mandal
Enter hours worked: 400
Enter hourly rate: 625
Employee: Ella Mae Mandal
Calculated Salary: 250000
```



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DATA STRUCTURE AND ALGORITHMS

10. MINI STUDENT INFORMATION SYSTEM

The screenshot displays the Visual Studio IDE with a C# project named 'MandalApp'. The code in 'Program.cs' prompts the user for student information and calculates the average grade and whether the student is 18 or older.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace MandalApp
{
    class Program
    {
        static void Main()
        {
            Console.WriteLine();
            Console.WriteLine("Enter Student Name:");
            string name = Console.ReadLine();
            Console.WriteLine("Enter Course:");
            string course = Console.ReadLine();
            Console.WriteLine("Enter Age:");
            int age = Convert.ToInt32(Console.ReadLine());
            Console.WriteLine("Enter Year Level:");
            string yearLevel = Console.ReadLine();
            Console.WriteLine("Enter Grade 1:");
            double grade1 = Convert.ToDouble(Console.ReadLine());
            Console.WriteLine("Enter Grade 2:");
            double grade2 = Convert.ToDouble(Console.ReadLine());
            Console.WriteLine("Enter Grade 3:");
            double grade3 = Convert.ToDouble(Console.ReadLine());
            double averageGrade = (grade1 + grade2 + grade3) / 3.0;
            bool isAdult = age >= 18;
            Console.WriteLine("\n--- STUDENT INFORMATION SUMMARY ---");
            Console.WriteLine($"Name: {name}");
            Console.WriteLine($"Course: {course}");
            Console.WriteLine($"Age: {age}");
            Console.WriteLine($"Year Level: {yearLevel}");
            Console.WriteLine($"Average Grade: {averageGrade:F2}");
            Console.WriteLine($"18 or Older: {isAdult}");
        }
    }
}
```

The output window shows the program's execution with the following input and output:

```
Enter Student Name: Ella Mae Mandal
Enter Course: BSIS
Enter Age: 20
Enter Year Level: 2
Enter Grade 1: 96
Enter Grade 2: 98
Enter Grade 3: 97

--- STUDENT INFORMATION SUMMARY ---
Name: Ella Mae Mandal
Course: BSIS
Age: 20
Year Level: 2
Average Grade: 97.00
18 or Older: True

C:\Csharp_workspace\example_app\MandalApp\MandalApp\bin\Debug\net10.0\Mandal
App.exe (process 20760) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Optio
ns->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```